

Lesson 4: Vector Analysis

Practical Session: Querying Attributes, Digitizing Vectors, and Clipping Rasters

Objectives

1. Understand how to query and edit the attribute table of an existing vector layer using Select by Expression and the Field Calculator.
2. Create a brand-new vector shapefile from scratch by digitizing a custom polygon directly on the map canvas.
3. Use a vector polygon as a mask layer to clip a raster dataset to a specific area of interest (AOI).

Preliminary Setup

1. Launch QGIS and initialise a new, empty project.
2. From your Lesson_1/Data folder, drag in `ne_10m_admin_0_countries.shp` and `ne_10m_populated_places_simple.shp`.
3. Also drag in the `HYP_50M_SR_W.tif` raster from Lesson_1/Data – you will clip this later in the session.
4. Create a new, empty folder named Lesson_4/Data on your computer. All new files you create today will be saved here.
5. Confirm the Project CRS in the bottom-right status bar reads EPSG:4326 (WGS 84).

Step 1: Querying and Calculating Attributes

Every vector feature is a row in a database. Before doing any geoprocessing, cartographers typically interrogate that database first – filtering for the features that matter and, where needed, calculating new values directly inside the table.

Procedure

1. Right-click `ne_10m_populated_places_simple` in the Layers Panel and choose Open Attribute Table. Briefly scroll across the columns and locate the fields **name** and **pop_max** (maximum population estimate).
2. Close the table. Right-click the layer again and choose Filter... (use Test instead of OK), or open Select features using an expression (the function-symbol icon in the Attributes Toolbar). Enter the expression: `"pop_max" > 5000000`
3. Click Select Features. The world's megacities are now highlighted in yellow on the map canvas.
4. With the megacities still selected, open the attribute table again and click the Show Selected Features button (bottom toolbar) to view only this subset.

5. Open the Field Calculator (abacus icon, top-left of the attribute table). Make sure Only update *n* selected feature(s) is *unchecked* this time, since the new field should apply to every city. Create a new field:
Output field name: pop_mill | Type: Decimal (real) | Expression: "pop_max" / 1000000.0
6. Click OK. A new column showing population in millions now appears for every populated place — calculated in one step rather than typed by hand.
7. Switch to the ne_10m_admin_0_countries layer. Open Select by Expression again and enter: "CONTINENT" = 'Europe'
8. With the European countries selected, right-click the layer → Export → Save Selected Features As.... Save it as a new Shapefile named europe_countries.shp inside Lesson_4/Data, keeping the CRS as EPSG:4326. Click OK.
9. Load the new layer if it is not added automatically and open its attribute table to confirm it contains only European countries. You have just created a new shapefile directly from an attribute query.

Note on Selections vs. New Files: *Selecting features only highlights them temporarily within the original layer – the source file on disk is untouched. To turn a selection into a reusable, independent dataset, you must explicitly export it, exactly as you did in step 7.*

Step 2: Digitizing a Custom Vector Layer

Not every vector layer comes from a downloaded dataset. Geologists frequently need to draw their own features – a fieldwork boundary, a sample area, or an outcrop polygon. QGIS lets you build a shapefile from nothing and populate it by hand.

Procedure

1. Go to Layer → Create Layer → New Shapefile Layer...
2. Set Geometry type to Polygon and confirm the CRS is EPSG:4326 – WGS 84.
3. Under New Field, add a text field named site_name (Type: Text data, length 50) and click Add to Fields List.
4. Click OK and save the file as study_area.shp inside Lesson_4/Data. An empty layer named study_area appears in your Layers Panel.
5. Select the study_area layer, then click the Toggle Editing icon (the pencil) in the Digitizing Toolbar to unlock it for editing.
6. Click Add Polygon Feature and, on the map canvas, click to place a vertex at each corner of an area covering several populated places near a country of your choice (for example, a box around three or four cities). Right-click to finish the shape.
7. A dialog box will prompt you for the attribute value. Enter a name for site_name (e.g. "My Study Area") and click OK.
8. Click Save Layer Edits (the floppy-disk-with-pencil icon), then click Toggle Editing again to lock the layer. Open the attribute table of study_area to confirm it now contains exactly one feature, with the geometry you drew and the attribute you typed.

Note on Digitizing Accuracy: *For this exercise, precision is not critical – any polygon roughly the size of a region is fine. In professional fieldwork, digitized boundaries are usually traced over a georeferenced basemap or satellite imagery (as in Lesson 3) to ensure spatial accuracy.*

Step 3: Clipping a Raster Using Your Vector Mask

A polygon can act as a cookie-cutter for raster data: any pixels outside its boundary are discarded, leaving only the data within your area of interest. This dramatically reduces file size and focuses analysis on the relevant extent – a step you will reuse in Lesson 5 when calculating vegetation indices.

Procedure

1. Open the Processing Toolbox (Processing → Toolbox).
2. Search for Clip Raster by Mask Layer (a GDAL tool) and double-click to open it.
3. Set Input layer to `HYP_50M_SR_W` (the Natural Earth hypsometric raster).
4. Set Mask layer to `study_area` – the polygon you digitized in Step 2.
5. Leave Source CRS and Target CRS at their defaults (EPSG:4326).
6. Under Clipped (mask), click the dropdown, choose Save to File..., and save the output as `hypso_clipped.tif` inside Lesson_4/Data. Click Run.
7. Once processing finishes, temporarily hide the original `HYP_50M_SR_W` layer. Notice that `hypso_clipped` shows colour and shaded-relief data only within the exact boundary of your `study_area` polygon – everything outside has been discarded.

Note on Vector vs. Raster Clipping: "Clip Raster by Mask Layer" resamples pixels to fit the mask boundary, similar in principle to the Warp tool you used in Lesson 2 – except here the target extent comes from a vector shape rather than a different CRS. If you instead need to clip one vector layer using another vector polygon as a cutter, the equivalent tool is Vector → Geoprocessing Tools → Clip.

Summary

1. **Attribute Tables Are Databases, Not Just Labels** – Select by Expression filters rows using logical queries (e.g. population thresholds, continent names), while the Field Calculator computes new columns from existing ones. Exporting a selection turns a temporary highlight into a permanent, independent file.
2. **Vector Layers Can Be Built, Not Just Downloaded** – New Shapefile Layer creates an empty container with a defined geometry type and schema; Toggle Editing and Add Feature then let you populate it by hand, exactly as a field geologist would digitize a mapped boundary.
3. **Vectors Can Constrain Rasters** – Clip Raster by Mask Layer uses any polygon – downloaded, selected, or digitized – to crop a raster to a precise area of interest, a foundational step for the raster analysis you will perform in Lesson 5.

Homework: Mapping a Buffer Zone Around a City

Objective

Combine attribute selection, buffering, and raster clipping into a single self-directed workflow, while reinforcing the CRS concepts from Lesson 2.

Procedure

1. Open the attribute table of `ne_10m_populated_places_simple` and use Select by Expression to find a single city of your choice (for example, "`name`" = 'Berlin').
2. Export the selected feature as a new shapefile named `selected_city.shp` in Lesson_4/Data.
3. Before buffering, reproject `selected_city` to a projected, metric coordinate system — for a European city, use EPSG:3035 (ETRS89-extended / LAEA Europe) from Lesson 2. Use Vector → Data Management Tools → Reproject Layer..., or simply save the layer with a new CRS via Export → Save Features As...
4. Run Vector → Geoprocessing Tools → Buffer on the reprojected point, setting Distance to 25000 (metres). Save the result as `city_buffer.shp`.
5. Use `city_buffer` as the mask layer in Clip Raster by Mask Layer to crop `HYP_50M_SR_W.tif`, exporting the result as `city_buffer_clip.tif`.
6. Build a Print Layout named Homework_4 containing the clipped raster, the buffer outline, the city point, a title, legend, scale bar, and north arrow. Export it as a PDF.

Submission Deliverable

Submit the exported PDF map layout, together with a short written answer (3–5 sentences) to the following question: *Why was it necessary to reproject the city point into a projected coordinate system (EPSG:3035) before creating a 25 km buffer, rather than buffering it directly in EPSG:4326? Refer back to what you learned about angular versus linear units in Lesson 2.*